All embedding-derived programs across observed and model contrasts
row order uses Ward clustering of the four standardized response columns for display only

A Complete program-response atlas

Observed 24h | Model 24h | Observed 1 week | Model 1 week | Genes 3–12 | Seed Jaccard | ORA label

P162 — Interferon Alpha Response
P013 — Interferon Alpha Response
P311 — Complement
P137 — Cell Cycle Checkpoints
P288 — Sensory Perception
P304 — Epithelial Mesenchymal Transition
P003 — G2M Checkpoint
P023 — Cell Cycle
P174 — E2F Targets
P045 — Cell Cycle

All 393 Louvain embedding programs
(only provisional pathway labels shown)

−0.6 −0.4 −0.2 0.0 0.2 0.4 0.6

Raw gene-program score difference (red = increased; blue = decreased)

○ Within-run program-effect FDR < 0.05